

# Application Note: On-the-Fly Electrostatic Potentials, Electric Fields, and Field Gradients for Microdroplet Chemistry in QUICK

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## Abstract

Local electrostatic observables provide a compact bridge between electronic structure and chemically interpretable molecular environments in microdroplets, sea-spray aerosols, and air-water interfaces. The electrostatic potential (ESP), electric field (EFIELD), and electric field-gradient tensor (EFG) are the scalar, vector, and tensor descriptors obtained from a single quantum-mechanical charge distribution at arbitrary probe points. Here we describe an application-oriented implementation in QUICK for evaluating ESP, EFIELD, and analytic EFG on user-supplied external grids, electron-density isosurfaces, and fixed probe sets along molecular trajectories. The target application is trajectory-resolved microdroplet chemistry, where interfacial charge separation, partial solvation, photochemical channels, and heterogeneous local fields have been proposed to control reactivity. The same post-SCF property engine is exposed through a no-I/O library API, allowing external molecular dynamics

drivers or trajectory-analysis tools to request ESP/EFIELD/EFG immediately after each converged single-point, geometry-optimization, or Born-Oppenheimer molecular dynamics step. The tensor convention is explicitly defined as  $E_i(\mathbf{C}) = -\partial V(\mathbf{C})/\partial C_i$  and  $G_{ij}(\mathbf{C}) = \partial E_i(\mathbf{C})/\partial C_j$ , the negative of the electrostatic-potential Hessian convention sometimes also denoted EFG. CPU serial and MPI implementations are available for ESP, EFIELD, and analytic EFG; the existing GPU ESP path defines the accelerator template for EFIELD and EFG kernels. File-based and no-I/O API evaluations agree to better than  $5 \times 10^{-9}$  a.u. for a water RHF/STO-3G validation grid, and analytic EFG agrees with a central finite-difference EFIELD reference to approximately  $10^{-9}$  a.u. in the current regression case. We outline an on-the-fly AIMD workflow for converting electronic trajectories into time series of local potentials, fields, field magnitudes, tensor traces, Frobenius norms, and anisotropies at chemically selected probes.

## 1 Introduction

Electrostatic observables are among the most direct ways to interpret a quantum-mechanical density in real space. The electrostatic potential reports the scalar potential experienced by a test charge, the electric field reports the directional force per unit charge, and the electric field gradient reports how rapidly that field changes locally. These quantities are useful for charge-model construction, field-sensitive reactivity, vibrational Stark interpretation, interfacial electrostatics, and polarizable embedding.

The motivating application for the present work is microdroplet chemistry. Aqueous microdroplets and sea-spray aerosols can display chemical behavior that differs strongly from bulk solution, including interfacial oxidation, photochemistry, and rate acceleration. Recent studies, including work by Francisco and co-workers on sea-spray microdroplet oxidation and hydrogen peroxide photolysis, have emphasized the possible roles of the air-water interface, charge separation, partial solvation, and heterogeneous local electric fields.<sup>1,2</sup> Recent reviews

and perspectives have likewise identified interfacial charge and local electrostatics as central, but not yet fully resolved, ingredients in microdroplet reactivity.<sup>3</sup> This creates a practical need for quantum-mechanical tools that can report not only energies and structures, but also the instantaneous scalar, vector, and tensor electrostatic environment experienced by reactive sites.

Trajectory-resolved electrostatics are especially useful because many chemical environments are intrinsically dynamic. Recent molecular simulation studies of interfacial and condensed-phase electrostatics have shown that analyzing local fields along trajectories can reveal heterogeneous microscopic environments that are obscured by structural averages alone.<sup>4</sup> For ab initio molecular dynamics (AIMD) or Born-Oppenheimer molecular dynamics (BOMD), the natural computational object is the converged electronic density at each frame. Once that density exists, evaluating ESP/EFIELD/EFG at selected probe points is a post-SCF property task, not a new electronic-structure problem.

This Application Note describes the current QUICK workflow for local electrostatic analysis along trajectories, with microdroplet and air-water-interface chemistry as the target use case. The implementation is staged: external-grid and electron-density-surface ESP/EFIELD/EFG are available in the standard input/output path; the same kernels are exposed through a no-I/O API for trajectory and driver use; and GPU ESP already exists as the accelerator template for future GPU EFIELD and EFG kernels. We focus on the application interface, sign conventions, validation protocol, and data products needed to use these observables in AIMD-style studies of reactive interfacial environments.

## 2 Definitions

For a probe point  $\mathbf{C}$ , nuclear positions  $\mathbf{R}_A$ , and electron density  $\rho(\mathbf{r})$ , the electrostatic potential is

$$V(\mathbf{C}) = \sum_A \frac{Z_A}{|\mathbf{C} - \mathbf{R}_A|} - \int \frac{\rho(\mathbf{r})}{|\mathbf{C} - \mathbf{r}|} d\mathbf{r}. \quad (1)$$

The electric field is

$$E_i(\mathbf{C}) = -\frac{\partial V(\mathbf{C})}{\partial C_i}, \quad (2)$$

and the field-gradient tensor reported by QUICK is

$$G_{ij}(\mathbf{C}) = \frac{\partial E_i(\mathbf{C})}{\partial C_j} = -\frac{\partial^2 V(\mathbf{C})}{\partial C_i \partial C_j}. \quad (3)$$

Equation 3 is the field-gradient convention. Some quantum-chemistry texts define EFG as the Hessian of the electrostatic potential itself; that convention differs by an overall sign.

In an AO basis, the electronic EFG contribution can be written as

$$G_{ij}^{\text{elec}}(\mathbf{C}) = \sum_{\mu\nu} P_{\mu\nu} \frac{\partial^2}{\partial C_i \partial C_j} (\mu | r_C^{-1} | \nu), \quad (4)$$

where  $P_{\mu\nu}$  is the density matrix and

$$(\mu | r_C^{-1} | \nu) = \int \chi_\mu(\mathbf{r}) \frac{1}{|\mathbf{r} - \mathbf{C}|} \chi_\nu(\mathbf{r}) d\mathbf{r}. \quad (5)$$

The analytic CPU EFG implementation evaluates these center derivatives by Obara–Saika-style recurrence over Gaussian basis functions.<sup>5</sup> A central finite-difference EFG from the analytic EFIELD remains available as a reference path.

### 3 Implementation in QUICK

QUICK now supports three complementary property routes:

1. File-based external-grid properties through the grid keywords for ESP, EFIELD, and EFG.
2. Electron-density isosurface properties through the corresponding surface keywords for ESP, EFIELD, and EFG.

3. A no-I/O API routine, `getQuickOEPROP`, that evaluates ESP, EFIELD, and/or EFG after a converged API energy or gradient call.

The no-I/O API has the form

```
call getQuickOEPROP(npoints, probe_xyz_bohr, ierr, &
                    esp=esp, efield=efield, efg=efg)
```

where the probe coordinates are in bohr and the optional output arrays are

```
esp(npoints)
efield(3,npoints)
efg(3,3,npoints)
```

The API delegates all property evaluation to the same OEPROP kernels used by the file-based routes. It returns a clean error when called before a converged density exists, when no output property is requested, or when the probe set is invalid.

The implementation uses the existing shell-pair decomposition in serial and MPI builds. For MPI, each rank accumulates its owned electronic shell-pair contribution and the electronic property arrays are reduced over  $N_{\text{grid}}$ ,  $3N_{\text{grid}}$ , or  $9N_{\text{grid}}$  double-precision values for ESP, EFIELD, or EFG, respectively. The direct nuclear terms are inexpensive and are evaluated on the host.

## 4 Trajectory and AIMD Workflow

The intended AIMD use pattern is post-SCF and frame local:

1. Converge the electronic structure for the current geometry.
2. Define probe points in a fixed laboratory frame, in a molecule-attached local frame, or on a generated density isosurface.
3. Evaluate ESP/EFIELD/EFG at those points using the file path or `getQuickOEPROP`.

4. Append compact descriptors to a trajectory CSV or binary analysis file.
5. Continue to the next geometry or dynamics step.

For externally generated trajectories, this workflow is trajectory post-processing. For an external MD or QM/MM driver linked against the QUICK API, the same call is on-the-fly because properties are requested immediately after each SCF or gradient evaluation. A native QUICK BOMD driver can use the same hook once the integration, restart, and energy conservation infrastructure is in place.

The validation trajectory driver writes one row per frame and probe point. The current CSV columns are:

```
frame,probe,energy_hartree,x_ang,y_ang,z_ang,esp,
efield_x,efield_y,efield_z,efield_mag,
efg_xx,efg_xy,efg_xz,efg_yx,efg_yy,efg_yz,
efg_zx,efg_zy,efg_zz,efg_trace,
efg_frobenius,efg_anisotropy
```

The tensor anisotropy is reported as the rotationally invariant deviatoric norm,

$$\eta_G = \left[ \frac{3}{2} \sum_{ij} \left( G_{ij} - \frac{\text{Tr}(G)}{3} \delta_{ij} \right)^2 \right]^{1/2}. \quad (6)$$

This descriptor is useful for detecting locally anisotropic electrostatic environments, while  $|\mathbf{E}|$ ,  $\text{Tr}(G)$ , and  $\|G\|_F$  provide complementary scalar summaries.

## 5 Validation

The present validation set covers file-based properties, no-I/O API properties, trajectory output, analytic EFG, and serial/MPI consistency. The no-I/O API was compared against the standard file-producing grid property route for ESP, EFIELD, and EFG. The test case is

water RHF/STO-3G with four fixed probe points. The first frame of a three-frame trajectory driver test is identical to the validation geometry, providing an additional trajectory-vs-API comparison.

Table 1: Current no-I/O OEPROP API validation for water RHF/STO-3G.

| Comparison                              | Values | Max. abs. dev. / a.u.  |
|-----------------------------------------|--------|------------------------|
| File ESP vs API ESP                     | 4      | $3.66 \times 10^{-11}$ |
| File EFIELD vs API EFIELD               | 12     | $4.71 \times 10^{-9}$  |
| File EFG vs API EFG                     | 36     | $3.85 \times 10^{-9}$  |
| Trajectory frame 1 ESP vs validation    | 4      | 0.0                    |
| Trajectory frame 1 EFIELD vs validation | 12     | 0.0                    |
| Trajectory frame 1 EFG vs validation    | 36     | 0.0                    |

Analytic EFG is also checked against the retained numerical EFG reference, which uses a central finite difference of EFIELD. In the current external grid regression case, analytic and numerical EFG agree to approximately  $10^{-9}$  a.u. Independent PySCF cross-code validation on external-grid and density-surface points gives maximum absolute differences of  $4.2 \times 10^{-6}$  a.u. or smaller across ESP, EFIELD, and EFG test sets, with stricter  $10^{-7}$ – $10^{-8}$  a.u. agreement for the water RHF/STO-3G surface cases.

## 6 GPU and Performance Considerations

The current implementation separates the property workflow from the SCF procedure: the density is converged once per frame, and ESP/EFIELD/EFG are then evaluated on the requested probe set. For a small number of probe points, the incremental cost is typically dominated by the SCF. For large external grids or dense molecular surfaces, the OEPROP contraction becomes a natural accelerator target.

QUICK already contains a CUDA/HIP electronic ESP path for external-grid OEPROP calculations. That path uploads grid coordinates and the converged density, evaluates the electronic attraction contribution on the accelerator, downloads the electronic property buffer, and adds the direct nuclear contribution on the host. The planned GPU EFIELD

and EFG kernels preserve the same host/device division, buffer ordering, and MPI reduction lengths as the CPU reference. The acceptance target for the accelerator extension is CPU/GPU agreement for ESP, EFIELD, analytic EFG, and the numerical EFG reference on the existing OEPROP regression suite.

## 7 Microdroplet-Chemistry Target

The immediate application target is compact local-electrostatics analysis of microdroplet, aerosol, and air-water-interface trajectories. In this setting, the quantities of interest are not only global dipoles or average surface potentials. The chemically relevant questions are local: what field does a peroxide O–O bond, thiol S–H bond, adsorbed ion, radical precursor, or surface hydroxyl group experience at the instant a reactive configuration appears? How anisotropic is the field around the reactive center? Does the local environment look bulk-like, interface-like, or charge-separated?

The present workflow is designed to answer these questions without claiming that QUICK is already a complete droplet dynamics package. One may generate AIMD or BOMD frames for finite water clusters, nanodroplets, slabs, or reactive solute-water aggregates, define chemically meaningful probe points, and then report time series such as

$$V(t), \quad |\mathbf{E}(t)|, \quad \text{Tr } G(t), \quad \|G(t)\|_F, \quad \eta_G(t).$$

For microdroplet chemistry, useful probes include bond midpoints, projected points along putative reaction coordinates, ion-solvation shell centers, density-isosurface points near adsorbed reactants, and surface-normal scan points that connect interior and interfacial regions. These observables can be correlated with bond distances, hydrogen-bond counts, proton-transfer coordinates, radical-pair separations, photochemical coordinates, solvent orientation, or vibrational frequency shifts. Because ESP, EFIELD, and EFG are evaluated from the same density and probe coordinates, they also provide a hierarchy of scalar, vector, and



tensor descriptors for testing field-driven mechanistic hypotheses in microdroplet chemistry and for constructing machine-learning or charge-model representations of reactive interfaces.

## 8 Limitations and Outlook

The present workflow should be described carefully. QUICK can evaluate ESP, EFIELD, and EFG after each converged energy or gradient call and exposes that capability through a no-I/O API. This is sufficient for trajectory post-processing and for on-the-fly property emission by an external driver. It is not yet a complete native AIMD package with thermostats, restartable dynamics, and automated OEPROP output at every step.

The next development steps are:

1. add production GPU EFIELD and EFG kernels using the existing GPU ESP implementation as the template;
2. benchmark OEPROP overhead on realistic AIMD probe sets, for example reactive water clusters, nanodroplet models, and interfacial slabs;
3. add native BOMD hooks that call `getQuickOEPROP` after each accepted step;
4. extend validation from small fixed water trajectories to finite-temperature AIMD trajectories of surface and interior water environments;
5. add small reactive microdroplet benchmarks, such as peroxide, thiol, halide, or proton-transfer motifs at finite water interfaces;
6. document sign, origin, and unit conventions in every trajectory output file.

These steps would turn the present validated property and API machinery into a publishable AIMD electrostatics workflow for local fields and field gradients in chemically heterogeneous environments, with microdroplet chemistry as the first application domain.

## 9 Data and Software Availability

The implementation is under development in QUICK. Validation inputs, API drivers, and trajectory-analysis scripts are included in the OEPROP API and GPU OEPROP validation directories of the development tree. TODO: add repository URL, commit hash, archived release DOI, and exact build instructions for the submitted version.

## Acknowledgement

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